

XMF-GNN: A cross-modality dynamic fusion heterogeneous graph neural network for network intrusion detection

Zhengxiang Ma^{a,c,d,*}, Yan Liu^a, Yu Chen^{a,b,c,d}, Zhibin Liu^a, Yinghao Li^e

^a School of Electronics and Information, Zhengzhou University of Aeronautics, Zhengzhou, 450046, China

^b Lantian Laboratory, Anyang, 455000, China

^c Henan Key Laboratory of General Aviation Technology, Zhengzhou University of Aeronautics, Zhengzhou, 450046, China

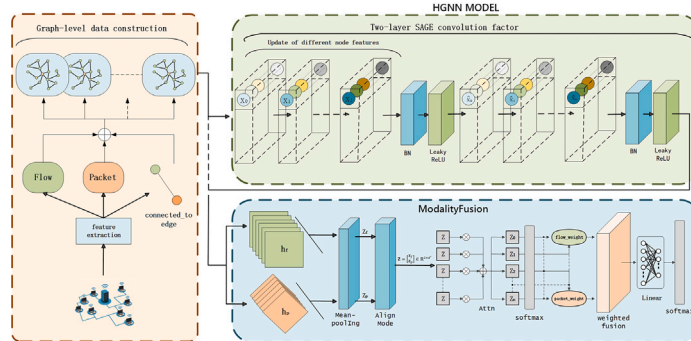
^d Collaborative Innovation Center of Aeronautics and Astronautics Electronic Information Technology, Zhengzhou, Henan Province, 450046, China

^e School of Cyber Science and Engineering, Zhengzhou University, Zhengzhou, 450000, China

HIGHLIGHTS

- A novel XMF-GNN framework fuses flow and packet data via dual-layer heterogeneous GNN.
- An attention-based fusion mechanism adaptively integrates multimodal network features.
- Extensive experiments validate XMF-GNN's advantages over existing fusion methods.

GRAPHICAL ABSTRACT



XMF-GNN detects network intrusions by converting flow and packet data into a heterogeneous graph, then utilizing HGNN and attention-based fusion to make accurate decisions.

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ABSTRACT

With the rapid development of the Internet, network attacks have become increasingly complex and covert. However, most existing intrusion detection methods focus on extracting static features from a single modality, making it difficult to fully exploit the complementary information across different modalities. Furthermore, these methods often lack structure-awareness and adaptive feature fusion capabilities when modeling complex attack behaviors, resulting in limited effectiveness in detecting diverse intrusion patterns. To address these challenges, we propose XMF-GNN, a heterogeneous graph neural network (HGNN) model enhanced with a cross-modality attention fusion mechanism. This model constructs a heterogeneous graph representation based on flow and packet modalities of network traffic, and employs an attention-driven modality fusion mechanism to dynamically learn the importance weights between modalities, enabling adaptive cross-modal feature fusion. The model architecture consists of a two-layer heterogeneous graph convolutional encoder and a modality attention module, which together effectively capture the intricate relationships between traffic flows and payload data in complex network environments. Experiments are conducted on two publicly available network intrusion detection datasets, CIC-IDS2017 and CIC-IoT2023. The results show that the proposed model achieves an F1 score of 0.977 on CIC-IDS2017 and an even higher F1 score of 0.985 on CIC-IoT2023 in multi-class classification tasks. It also

* Corresponding author at: School of Electronics and Information, Zhengzhou University of Aeronautics, Zhengzhou, 450046, China.

Email addresses: mzx@zua.edu.cn (Z. Ma), ly2023@zua.edu.cn (Y. Liu), chenyu@zua.edu.cn (Y. Chen), lzb2023@zua.edu.cn (Z. Liu), yinghaoli@zzu.edu.cn (Y. Li).

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significantly outperforms state-of-the-art methods in terms of accuracy and other mainstream evaluation metrics, demonstrating the effectiveness and robustness of the proposed cross-modality fusion mechanism in handling complex intrusion detection scenarios.

1. Introduction

Intrusion detection refers to the analysis of network data to determine whether network activities exhibit malicious intent [20]. In recent years, machine learning (ML) has demonstrated powerful modeling and predictive capabilities across various domains, contributing significantly to the advancement of modern agriculture, industry, and healthcare [14–16,19]. These advancements provide valuable technical inspiration and methodological insights for the development of network intrusion detection systems (NIDS).

ML-based Network Intrusion Detection Systems primarily rely on the analysis of numerical features derived from flow-level metadata, following a flow-based detection paradigm [3,5,18]. These methods typically model behavioral patterns of network communication—such as data volume transferred between endpoints, connection duration, and interaction frequency—to identify anomalous activities. For example, anomaly-based intrusion detection systems (AIDS) utilize such strategies and are particularly effective at detecting novel communication anomalies, including port scanning, denial-of-service (DoS/DDoS) attacks, and zero-day attacks [10,12]. However, since AIDS relies solely on statistical behavioral features, it may fail to detect more subtle attacks hidden within the payload of a single packet [27].

On the other hand, with the rapid advancement of deep learning and natural language processing (NLP) [2], some studies have begun representing and modeling packet payloads in numerical or textual formats, leading to the emergence of packet-based detection architectures [1,13]. This approach overcomes the limitations of traditional flow-level detection by enabling in-depth analysis of packet content and identifying finer-grained attack patterns. A typical example is the signature-based intrusion detection system (SIDS), which identifies potential threats by matching packet content with known attack signatures or feature libraries. This method is particularly effective in detecting well-defined attacks such as SQL injection, cross-site scripting (XSS), or buffer overflows. Nevertheless, due to its dependence on static feature databases, it remains vulnerable to novel or obfuscated attacks [29].

Although both detection paradigms have achieved notable progress, each presents inherent limitations. Flow-based methods, which focus on statistical indicators such as source/destination IP addresses, port numbers, communication duration, and packet rate, are effective in identifying attacks characterized by conspicuous traffic anomalies, such as port scanning or DDoS. However, they often overlook content-level details and struggle to detect fine-grained attacks embedded in single packets, such as SQL injection or malicious script injection. Conversely, packet-based methods analyze packet payloads to identify attacks with specific content patterns, and are therefore well suited for detecting content-related threats like buffer overflows or remote code execution [6]. Yet, these approaches often encounter performance bottlenecks when processing high-frequency, large-scale network traffic and lack the capability to perceive global behavioral patterns, making it difficult to detect complex attacks involving evolving behavioral patterns. Consequently, building an efficient and robust NIDS is crucial for ensuring sustained cybersecurity.

In recent years, an increasing number of studies have focused on effectively integrating the macroscopic perception capability of flow-level modeling with the microscopic recognition strength of packet-level semantic analysis [7,17,23]. Such hybrid approaches are expected to be key in improving the accuracy, robustness, and generalization capability of intrusion detection systems. Ideally, an NIDS architecture should be capable of simultaneously perceiving high-level behavioral patterns and low-level content features, dynamically balancing the importance of

both types of information, and adaptively adjusting detection strategies according to the nature of the attack.

To address these challenges, this paper proposes a novel cross-modality fusion HGNN model. This approach integrates both flow-level and packet-level information into a unified representation using a heterogeneous graph structure. To overcome the limitations of traditional methods in capturing both local information and global behavior, XMF-GNN introduces an innovative multimodal data fusion strategy that embeds flow- and packet-level features into a heterogeneous graph, enabling joint modeling and inference through a Graph Neural Network (GNN). Moreover, an attention-driven modality fusion mechanism is employed to dynamically learn the relative importance between modalities, facilitating adaptive cross-modal feature fusion. This dynamic integration empowers the system to detect a broader range of network attacks, thereby enhancing overall detection accuracy.

The main contributions of this work are summarized as follows:

- We propose a novel framework named XMF-GNN, which fuses flow-level and packet-level data and employs a dual-layer SAGE-based heterogeneous GNN encoder for graph-level classification. This design enables timely aggregation and holistic processing of heterogeneous network traffic data.
- We introduce an attention-driven cross-modality fusion mechanism that dynamically learns the importance weights between different modalities, achieving adaptive integration of multimodal features.
- We conduct a comprehensive quantitative comparison between our proposed model and other state-of-the-art cross-modality fusion methods to demonstrate its effectiveness.

The remainder of this paper is organized as follows: [Section 2](#) reviews related work on GNN applications in NIDS. [Section 3](#) details the components of the proposed XMF-GNN model. [Section 4](#) describes the experimental setup and presents the evaluation results. [Section 5](#) concludes the paper and outlines potential directions for future research.

2. Related work

This section provides a detailed review of existing research on the application of Graph Neural Networks (GNNs) in the development of NIDS. It is noteworthy that most current NIDS approaches focus on modeling static features of nodes or edges, while overlooking the semantic diversity and interaction relationships inherent in multimodal network traffic data. These methods typically convert traffic into a single-modality graph structure, where nodes represent IP addresses, hosts, or sessions, and edges reflect communication or connection relationships. Under such designs, models tend to emphasize behavioral-level statistical features while lacking the capability to perceive fine-grained content-level attributes such as packet payloads and protocol semantics. As a result, their effectiveness is limited when facing content-based attacks, such as malicious code injection or stealthy infiltration.

As an emerging paradigm in network security analysis, GNNs have gained increasing attention in NIDS research. Their core detection mechanism relies on iterative feature propagation and topology-aware aggregation to jointly model traffic behavior and communication structure, enabling the generation of discriminative graph representations for threat detection.

2.1. Intrusion detection based on single-modality GNNs

With the widespread application of GNNs in the field of cybersecurity, researchers have increasingly explored their potential in NIDS. One line of work focuses on modeling network traffic from a single

modality by abstracting hosts, connections, or sessions in the network as nodes, and constructing graph data through topological structures and statistical attributes to improve the accuracy and generalization of attack detection. These approaches typically build graphs based on flow-level structural information or static features (such as IP addresses, ports, traffic volume, etc.), leverage graph embedding techniques to learn latent interaction patterns, and utilize models like Graph Convolutional Networks (GCNs) and Graph Attention Networks (GATs) for classification. Such methods are relatively simple in structure and computationally efficient, and they perform reasonably well in detecting conventional types of attacks [5,11].

Lo et al. [18] proposed the E-GraphSAGE model, which performs fine-grained traffic recognition through edge-level classification. The framework extends the classic GraphSAGE algorithm [11] by encoding network traffic structure information into a heterogeneous graph representation, effectively capturing edge features and topological dependencies in IoT environments. However, this model faces scalability issues due to its computational complexity when processing complete traffic graphs, resulting in limited real-time inference capabilities.

To address these issues, Chang and Branco [5] proposed two key improvements: a residual learning mechanism to mitigate class imbalance—particularly to enhance the detection of minority attack classes—and the design of an edge-based residual graph attention network (E-ResGAT), which improves model efficiency through dynamic attention-based weighting. Nonetheless, the method still relies on fixed neighborhood sampling and static attention mechanisms, making it less adaptive to dynamic network environments.

Xiao et al. [31] introduced a hierarchical graph embedding-based intrusion detection approach. This study constructed a dual-granularity graph model by transforming traffic into first-order and second-order graphs. The first-order graph takes individual hosts as analysis units and learns host-level behavior features through IP and port topologies. The second-order graph uses quadruples (source IP, source port, destination IP, and destination port) to build communication topologies that capture interaction features at the network level. By combining graph embeddings with raw traffic features and applying a random forest classifier, the model supports multiclass attack detection.

Altaf et al. [3] further proposed an enhanced GNN-based NIDS framework that represents IoT sessions as nodes (using IPs and ports) and encodes flow-level statistical features (e.g., flow size, protocol type) as edge attributes. This approach effectively detects both network-layer (e.g., DDoS, port scanning) and application-layer attacks. However, its feature representation is limited in dimensionality, and attacks relying heavily on payload content can still evade detection.

GNN-based methods that rely on a single modality are effective in modeling structural or behavioral patterns within network flows and have demonstrated strong performance in detecting attacks with clear communication characteristics, such as DDoS and port scanning. However, these methods often lack the capability to model packet payloads and semantic contextual information, which limits their effectiveness when facing more complex and dynamic application-layer attacks [24].

2.2. Intrusion detection based on multimodal GNNs

With the increasing diversity and stealthiness of network attacks, single-modality features are no longer sufficient to support effective threat modeling in intrusion detection. To address this limitation, researchers have proposed multimodal GNN approaches that integrate various dimensions of network traffic, aiming to combine information from multiple dimensions for more comprehensive and adaptive intrusion detection [8,23].

Premkumar et al. [23] proposed a multimodal NIDS framework based on Graph Representation Learning (GRL). The method first uses GNNs to generate packet-level embeddings, then concatenates these vectors with traditional flow-level features for final classification. While this design

captures both detailed packet content and broad traffic features, it suffers from limitations due to its two-stage fusion strategy. Specifically, cross-modal interactions occur only at the final classification stage, lacking direct semantic alignment at the graph structure level. Moreover, its multi-step processing increases computational complexity, making it less suitable for real-time detection scenarios.

Building upon this, Farrukh et al. [8] introduced the XG-NID framework, a novel HGNN designed for end-to-end fusion of flow and packet data. This work constructs a dual-modality heterogeneous graph topology—flow-nodes encoding NetFlow features and packet-nodes embedding Deep Packet Inspection (DPI) information. The framework pioneers the use of two edge types—“contain” and “link”—to associate protocol-layer and payload-layer semantics. A modality-aware mechanism is also designed to enable interactions between modalities. Compared to [23], this approach resolves issues related to payload information neglect but still relies on static modality fusion and lacks fine-grained semantic awareness.

In summary, although existing GNN-based NIDS models have achieved certain success in node- or edge-level classification tasks, they are generally limited to single-modality modeling. Most approaches focus on local features and lack the ability to integrate heterogeneous sources of information, failing to capture global relationships and semantic diversity in network traffic. While some recent studies attempt to use multimodal data fusion to address these limitations, issues such as staged feature interaction and static fusion still persist.

To address these gaps, our proposed approach introduces a modality-attentive mechanism that performs adaptive fusion of behavioral and content features within a graph-level classification framework. Compared to traditional methods, our model not only excels at detecting content-based attacks with semantic patterns but also demonstrates strong capabilities in identifying large-scale traffic anomalies and behavior-based threats. The graph-level classification paradigm further enhances the model's capacity to understand complex structural relationships in network communications, thereby improving its accuracy and robustness in handling diverse network threats.

3. Proposed model

In this section, we provide a detailed explanation of the implementation of the proposed Cross-Modality HGNN. The core of our model lies in introducing a modality attention mechanism, which enables dynamic and adaptive fusion of flow-level and packet-level information in graph-level tasks. This approach significantly enhances the model's ability to discriminate between different types of attacks. The framework integrates flow and packet information using HGNN, leveraging the structure of adjacent nodes to achieve comprehensive feature aggregation.

The proposed XMF-GNN framework extracts flow-level and packet-level features from raw network traffic data and models them through an HGNN. Crucially, a modality attention mechanism is used to dynamically fuse the two types of modality information—flow and packet. This dynamic fusion enables the model to learn modality-specific importance weights in an adaptive manner, equipping the model with autonomous “judgment capability” for more comprehensive behavioral analysis in network intrusion detection. The overall architecture of the model is illustrated in Fig. 1. This dynamic fusion mechanism effectively addresses the issue of modality dependence variance across different attack types, and enhances the model's ability to capture complex interactions in network traffic, resulting in more accurate and effective intrusion detection.

Initially, the model extracts relevant modality-specific information from raw traffic data and applies the heterogeneous graph construction method proposed by Farrukh et al. [8] in the XG-NID framework to build the graph structure. The constructed graph is then fed into our designed heterogeneous GNN for information propagation and aggregation. We perform graph-level pooling on the updated node features

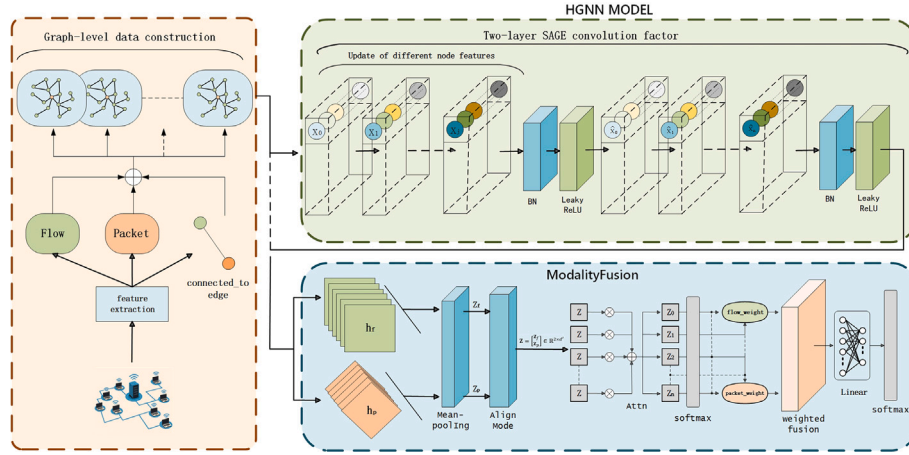


Fig. 1. The system first extracts two modalities of features—Flow and Packet—from raw network traffic data. These features are then used to construct graph-level representations, forming a heterogeneous graph structure. The heterogeneous graph is input into the HGNN model for graph neural inference. Within HGNN, the features of different node types are updated to obtain two distinct graph-level representations, h_f and h_p . These representations are passed through an average pooling layer and a linear transformation to project them into a shared semantic space. The aligned features are concatenated into a unified representation vector Z . A fusion module combined with an attention mechanism is used to learn the importance of each modality. Finally, the integrated features are passed through a linear layer to produce the final output.

to obtain graph-level representations for both flow and packet modalities. These two graph representations are then passed into the modality fusion module, where the attention mechanism is used to learn the relative importance of each modality. This process allows the model to dynamically adjust its reliance on each information source, thereby assigning more effective weights depending on the attack type.

3.1. Feature extraction

To better distinguish packet-level information from its corresponding flow-level context, and to ensure each flow contains meaningful payload information derived from its associated packets, we employ a layered feature extraction strategy. During feature generation, we utilize the NFStream framework [4] to analyze each flow and its constituent packets. The module extracts a total of 76 flow-level features, including duration, number of transmitted bytes, protocol type, communication frequency, and so on, as well as 14 packet-level features, with an emphasis on protocol fields and payload distribution content. To improve the system's ability to detect content-based attacks, we transform each packet's payload into a fixed-length feature vector of 1500 dimensions. This transformation method follows the approach proposed by Farrukh et al. [6]: each byte in the payload, expressed in hexadecimal, is mapped to an integer between 0 and 255, forming a dense numerical vector. If the packet's length is less than 1500 bytes, zero-padding is applied; if a packet contains no payload, the entire vector is set to zero. This resulting vector is used as the attribute of each Packet node during the graph construction phase and is fed into the model for subsequent learning.

We enhanced the sliding window-based time-domain statistical feature extraction method originally proposed by Farrukh et al. [7]. The specific procedure is detailed in Algorithm 1. A sliding window refers to continuously computing statistics over a predefined temporal range—for example, the number of packets within the last 60 s or the proportion of ICMP packets in the past 30 s. Such features help capture whether communication behaviors become abruptly dense or sparse, whether certain types of packets (e.g., TCP SYN) repeatedly occur within a short time span, or whether temporal attack patterns such as slow attacks or scanning activities are present.

However, the original single-scale sliding window uses a fixed time granularity, which cannot adapt to the diverse temporal patterns of different attacks. For instance, a scan may last for several minutes, whereas a brute-force attack might complete within 10 s. Fixed time

Algorithm 1: Multi-scale explainable feature extractor for NIDS.

Input: $\mathcal{W} = \{w_1, \dots, w_n\}$: Set of sliding window sizes (e.g., 10 s, 30 s, ...)

$\mathcal{F}$: Conventional flow-level features

Output: $\mathcal{E}$: Multi-scale enriched features for NIDS

1 Step 1: Initialize Multi-Scale Buffers

2 for each destination D_j do

3 for each window size $w \in \mathcal{W}$ do

4 Initialize buffer: $W_j[w] \leftarrow [0, 0, \dots, 0]$ (length = w);

5 Step 2: Update Buffers and Compute Features

6 for each time step t_i do

7 for each destination D_j do

8 for each window size for $w \in \mathcal{W}$ do

9 do

10 Update $W_j[w]$ with packets received at t_i ;

11 Compute temporal features $T_j[w]$, such as;

- 12 - Packet rate (pkt/s);
- 13 - Average packet size;
- 14 - TCP SYN ratio;
- 15 - Directional imbalance (src/dst ratio);
- 16 - ICMP proportion, RTT variance, etc.;

17 Step 3: Combine Multi-Scale Features

18 for each destination D_j do

19 Concatenate: $\mathcal{E}_j \leftarrow \text{Combine}(\mathcal{F}, T_j[w_1], \dots, T_j[w_n])$;

20 return $\mathcal{E} = \{\mathcal{E}_1, \mathcal{E}_2, \dots, \mathcal{E}_m\}$

scales may be too coarse, and varying connection durations make static window sizes difficult to generalize. To address these limitations, we propose employing multiple time scales simultaneously, computing behavioral features across several sliding windows in parallel. This approach provides the model with more fine-grained temporal context.

Specifically, for each network flow, a set of statistical features is computed in parallel across multiple predefined time windows. These features reflect the traffic patterns and communication behaviors of a connection under different temporal resolutions, thereby enhancing the model's ability to distinguish between various intrusion types,

such as short-lived burst attacks and long-term scan activities. The extracted features from each time window are concatenated to form a comprehensive vector, which is used as the feature input for the Flow node: " $FlowFeature$ " = $[F_{10s}, F_{30s}, F_{60s}, F_{300s}] \in \mathbb{R}^n$, where each F_{ts} represents the subset of features computed within a specific time window.

These temporal features are systematically updated over time and combined with traditional flow-based features. The resulting metrics provide a dynamically updating output framework for intrusion detection. Temporal features are periodically recalculated using multi-scale sliding windows, ensuring the feature space remains temporally relevant and improves predictive accuracy. This integration of time-based and flow-based features enables a more dynamic understanding of network behavior, which is crucial for identifying and interpreting complex, time-sensitive attacks.

3.2. Graph-level construction

To effectively utilize the extracted flow-level and packet-level features, we transform these two modalities into a heterogeneous graph structure, which serves as the input to our network model. To comprehensively analyze network traffic for intrusion detection, we follow the graph construction approach proposed by Farrukh et al. [7], which involves building two types of nodes and using two edge types—Link Edges and Contain Edges. The Link Edge incorporates a Time Delta feature that captures the inter-packet time intervals within a flow, indicating periods of activity or idleness. In contrast, the Contain Edge connects each flow node to its associated packet nodes and includes a 4-dimensional feature vector describing edge directionality and protocol-layer sizes.

However, we observed that node feature updates depend primarily on the node attributes themselves, rather than on the specific edge types. Moreover, incorporating multiple edge types significantly increases computational redundancy in the graph convolution operations, thereby increasing the model's complexity. To address this, we eliminate the explicit representation of both edge types and instead retain only the connection relationships between the two types of nodes. The original edge features are embedded directly into the corresponding node attributes. The construction process is illustrated in Fig. 2. This approach enables richer contextual information to be embedded within node representations, thereby enhancing the model's capacity to classify attack types accurately. In particular, the deep modeling of interactions between flow and packet data allows the model to more precisely capture network behaviors.

We denote the resulting heterogeneous graph as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the node set and $\mathcal{E}$ is the edge set. The nodes are categorized into two types: flow nodes V_f and packet nodes V_p . Each flow node $v_f \in V_f$ represents a network flow F , whose feature vector $\mathbf{h}_{v_f} \in \mathbb{R}^d$ is composed of 76 flow-level features and 52 temporal features generated by the multi-scale sliding window technique. Each packet node $v_p \in V_p$ corresponds to a data packet P_i , with a feature vector $\mathbf{h}_{v_p} \in \mathbb{R}^{1500}$, comprising a 1500-dimensional payload encoding and 14 protocol-level header features.

The original edge features are integrated into the corresponding flow and packet node features. As a result, the formal edge definitions for containment and link relationships are no longer explicitly represented. Instead, feature propagation is achieved via node-level aggregation. The following aggregation functions ψ represent how node features are updated using previously defined edge types, now implicitly encoded within node attributes:

$$v'_f = \psi(v_f, \{\mathcal{E}_c(v_f, v_{pi}) \mid 1 \leq i \leq n\}) \quad (1)$$

$$v'_p = \psi(v_p, \{\mathcal{E}_l(v_{pi}, v_{pi+1}) \mid 1 \leq i < n\}) \quad (2)$$

Here, the updated node representations $v'_f \in \mathcal{V}_f$, $v'_p \in \mathcal{V}_p$ are connected via a unified connected-to edge type, denoted as $\mathcal{E}_{f,p}$, simplifying the overall graph structure while preserving essential relational

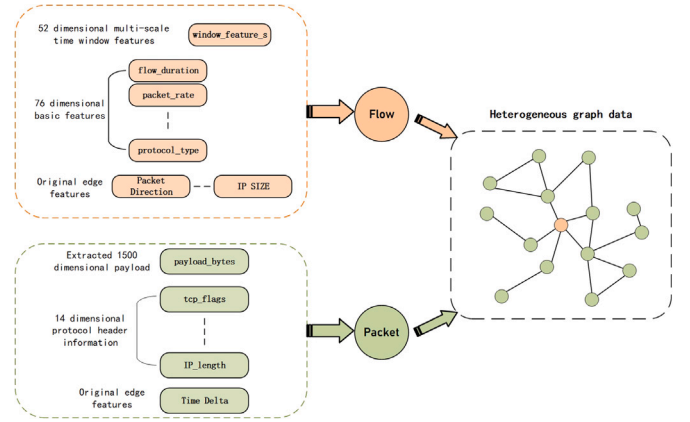


Fig. 2. Heterogeneous graph construction pipeline.

information. The resulting node set $\mathcal{V}$ is formally defined as:

$$\mathcal{V} = (\mathcal{V}_f \cup \mathcal{V}_p, \mathcal{E}_{f,p}) \quad (3)$$

3.3. Heterogeneous graph neural network

The HGNN model is designed to leverage the heterogeneous graph structure, enabling deeper interactions between the microscopic packet-level features and the macroscopic flow-level features embedded within network traffic data. Our model builds upon the graph attention convolution operator SAGEConv proposed by Hamilton et al. [11]. The GraphSAGE framework effectively supports feature aggregation across heterogeneous nodes by leveraging neighborhood sampling and feature propagation mechanisms. This allows for the dynamic integration of local features from different node types, thereby capturing the complex relationships among diverse node and edge types in network traffic graphs. The algorithmic process of the SAGEConv operator is illustrated in Fig. 3. Taking flow nodes as an example, SAGEConv performs neighborhood sampling over their connected packet nodes, aggregates the features from the sampled packet neighbors, and updates the flow node embedding via a mean aggregator.

We describe our model layer-wise as follows. The mathematical expression is shown in formula (4):

$$h_{v_i}^{(1)} = \sigma \left(\text{SAGEConv} \left(h_{v_i}^{(0)}, \mathbf{A}, \mathbf{E} \right) \right) \quad (4)$$

Here, $h_{v_i}^{(0)}$ denotes the initial feature of node type v_i , $\mathbf{A}$ is the adjacency matrix, and $\mathbf{E}$ represents the connected-to edge set. The activation function $\sigma(\cdot)$ used in this model is LeakyReLU, as defined in Eq. (5). This function improves model expressiveness by allowing a small gradient to flow when the input is negative, typically with $\alpha = 0.01$, making it more "tolerant" to negative values compared to standard ReLU.

$$\text{LeakyReLU}(x) = \begin{cases} x & \text{if } x \geq 0 \\ \alpha x & \text{if } x < 0 \end{cases} \quad (5)$$

Each SAGEConv layer performs feature aggregation and updating across the graph structure, enabling the model to capture deeper dependencies among nodes. We employ a two-layer graph convolutional architecture. The node feature update rule at layer $h_{v_i}^{(l)}$ is defined by Eq. (6):

$$\mathbf{h}_{v_i}^{(l)} = \sigma \left(\mathbf{W}_\alpha \cdot \mathbf{h}_{v_i}^{(l-1)} + \frac{1}{|\mathcal{N}(v_i)|} \sum_{v_j \in \mathcal{N}(v_i)} \mathbf{W}_\beta \cdot \mathbf{h}_{v_j}^{(l-1)} \right) \quad (6)$$

Here, $v_i \in V$ is any node in the graph, $\mathcal{N}(v_i)$ denotes the set of its neighboring nodes defined by edges E , and $\mathbf{W}_\alpha, \mathbf{W}_\beta$ are the learnable weights for self-loop and neighbor aggregation, respectively. These

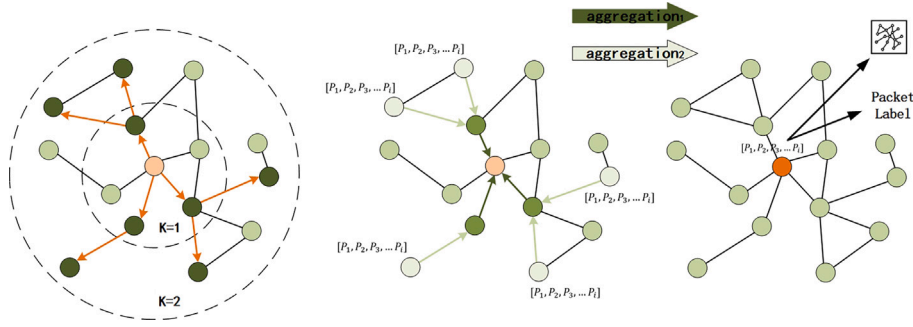


Fig. 3. Algorithm flow of SAGEConv operator.

ensure that each node retains its own information while incorporating contextual features from its neighbors. As shown in Fig. 3, flow nodes preserve their statistical properties through W_a , while learning interaction patterns from packet neighbors. Conversely, packet nodes retain their payload semantics while becoming aware of interaction frequency with flow nodes.

To mitigate the issues of gradient explosion or vanishing, we apply Batch Normalization (BN) independently to each node type, which standardizes node embeddings and ensures stability across different node distributions. The BN operation is given by Eq. (7):

$$\text{BN}(x) = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}} \cdot \gamma + \beta \quad (7)$$

Global pooling serves as a crucial step to transform node-level representations into graph-level embeddings, a common practice in graph classification tasks. It aggregates all node embeddings in a graph into a single vector and plays a key role before the dynamic weight fusion mechanism. Let G_k represent the k graph, with $V_f^{(k)}$ and $V_p^{(k)}$ denoting the flow and packet node sets, respectively. After the second convolutional layer, each node embedding $h_{v_i}^{(2)} \in \mathbb{R}^d$, and the graph-level embeddings for the flow and packet nodes are computed as follows:

$$\mathbf{h}_f^{(k)} = \frac{1}{|V_f^{(k)}|} \sum_{v_i \in V_f^{(k)}} \mathbf{h}_{v_i}^{(2)} \quad (8)$$

$$\mathbf{h}_p^{(k)} = \frac{1}{|V_p^{(k)}|} \sum_{v_i \in V_p^{(k)}} \mathbf{h}_{v_i}^{(2)} \quad (9)$$

3.4. Attention-driven modality dynamic fusion mechanism

After multiple rounds of message passing and aggregation via the HGNN, the model obtains graph-level embeddings for both Flow and Packet node types. These two embeddings respectively capture statistical behavioral features and semantic content features. Most previous works statically encode the fusion of Flow and Packet modalities during the graph construction phase, resulting in a fixed fusion ratio. However, such a rigid strategy fails to adaptively balance the importance of different modalities based on the attack characteristics of each sample. Our experiments reveal that different attack types rely on different modalities. For instance, in Brute Force attacks, the attacker often submits numerous login attempts in a short time, and the login requests typically reside in the payloads within the Packet modality, where the username and password fields are embedded. To address this issue, we propose the ModalityFusion module, which aims to learn an adaptive attention mechanism that assigns appropriate weights to each modality and generates a unified fused graph representation for downstream classification tasks.

A prerequisite for applying attention is mapping the two modalities' embeddings into the same hidden space. Since the graph-level embeddings $\mathbf{h}_f$ and $\mathbf{h}_p$ obtained via global average pooling, originate from

different data modalities, their original representations are not directly comparable. Therefore, we employ linear transformations to project the two modalities into a shared semantic space, facilitating comparability for subsequent attention weighting. Using trainable projection matrices W_f , W_p and bias terms $\mathbf{b}_{f,p} \in \mathbb{R}^{d'}$, we align both embeddings into the same dimensional space and distribution. The projected results are stacked and concatenated into a 3D tensor, forming a $2 \times d'$ modality representation set $\mathbf{Z}$ for each batch. The projection is defined in Eq. (10):

$$\mathbf{z}_f = W_f \mathbf{h}_f + \mathbf{b}_f, \mathbf{z}_p = W_p \mathbf{h}_p + \mathbf{b}_p, \mathbf{Z} = \begin{bmatrix} \mathbf{z}_f \\ \mathbf{z}_p \end{bmatrix} \in \mathbb{R}^{2 \times d'} \quad (10)$$

We adopt a shared attention mechanism, meaning a single parameter matrix W_a is used to score both modality representations. This design avoids introducing human bias and ensures that the attention reflects the relative importance of each modality.

$$\alpha = \text{softmax}(W_a \cdot \tanh(\mathbf{Z})) \in \mathbb{R}^{2 \times 1} \quad (11)$$

To enhance discriminability, a non-linear mapping via the tanh function is used, followed by softmax normalization to yield the probability weights for the flow and packet modalities. Finally, based on the output attention weights, the model performs a weighted combination of the two modalities to obtain the final graph-level representation $\mathbf{h}_\zeta$. This representation preserves information from both modalities while automatically highlighting the more critical modality through learnable attention weights. The computation is defined in Eq. (12):

$$\alpha_f = \alpha_0, \alpha_p = \alpha_1 \quad \mathbf{h}_\zeta = \alpha_f \cdot \mathbf{h}_f + \alpha_p \cdot \mathbf{h}_p \quad (12)$$

The fused graph representation $\mathbf{h}_\zeta$ is then passed through a series of fully connected layers to produce the final classification output. These layers progressively reduce the embedding dimension while refining the learned features:

$$\text{Out} = \text{LogSoftmax} (W_2 \cdot \text{ReLU} (W_1 \cdot \text{ReLU}(W_0 \cdot \mathbf{h}_\zeta))) \quad (13)$$

Here, $W_0 W_1$ and W_2 are the weight matrices of the fully connected layers, and the LogSoftmax function converts the final output into class probabilities. In summary, the proposed HGNN model effectively integrates packet-level and flow-level information within a heterogeneous graph framework. It performs hierarchical message aggregation and representation learning across different graph structures, capturing deeper interaction patterns. Meanwhile, the attention-based dynamic fusion mechanism enables adaptive modeling of modality preferences across different attack types, significantly improving overall classification accuracy. The detailed procedure is outlined in Algorithm 2.

Algorithm 2: Modality attention fusion mechanism.

Input: $\mathbf{h}_f$: Flow modality embedding;
 $\mathbf{h}_p$: Packet modality embedding;
 $\mathbf{W}_f, \mathbf{W}_p$: Projection weights;
 $\mathbf{W}_a$: Attention weights

Output: $\mathbf{h}_\zeta$: Fused multimodal representation

```

1  $\mathbf{z}_f \leftarrow \mathbf{W}_f \cdot \mathbf{h}_f + \mathbf{b}_f$ ;
2  $\mathbf{z}_p \leftarrow \mathbf{W}_p \cdot \mathbf{h}_p + \mathbf{b}_p$ ;
3 for each sample  $i$  in batch do
4    $\mathbf{Z}_i \leftarrow \text{concat}(\mathbf{z}_f[i], \mathbf{z}_p[i])$ ;
5    $\alpha_i \leftarrow \text{softmax}(\mathbf{W}_a \cdot \tanh(\mathbf{Z}_i))$ ;
6    $\alpha_f \leftarrow \alpha_i[0], \alpha_p \leftarrow \alpha_i[1]$ ;
7    $\mathbf{h}_\zeta \leftarrow \alpha_f \cdot \mathbf{h}_f[i] + \alpha_p \cdot \mathbf{h}_p[i]$ ;
8 return  $\mathbf{h}_\zeta$ ;

```

4. Experiments

This section first describes the dataset, evaluation metrics, experimental setup, and baseline methods used in the experiment. We will conduct a comprehensive evaluation of the results of the proposed model, comparing its effectiveness with three widely used machine learning models and five state-of-the-art baseline methods. Subsequently, ablation experiments were conducted to validate the effectiveness of the proposed improvements in this study, and we delve into how the model dynamically models the dependence of different attack samples on flow and packet modalities through attention mechanisms, enabling adaptive learning of modal preferences. Additionally, we performed importance analysis of the model's input features using the Integrated Gradients method [28].

4.1. Dataset

To ensure a robust and comprehensive evaluation of the XMF-GNN framework proposed in this paper, we adopted two widely recognized benchmark datasets: CIC-IoT2023 [21] and CIC-IDS2017 [25]. Although we have already detailed the preprocessing workflow for CIC-IoT2023, we will not repeat the similar steps for CIC-IDS2017 here to avoid redundancy and maintain readability. Both datasets were processed using a consistent pipeline to ensure fairness in comparison.

4.1.1. CIC-IDS2017 dataset

The CIC-IDS2017 dataset was jointly developed by the Canadian Communications Security Establishment (CSE) and the Canadian Institute for Cybersecurity (CIC) to simulate benign traffic and complex attacks in a realistic network environment. Collected from July 3 to July 7, 2017, the dataset consists of five days of network traffic generated by 25 users, totaling approximately 48.8 GB of data. It is provided in both PCAP and CSV formats. The CSV files contain 80 flow features extracted using CICFlowMeter. The dataset includes attack categories such as Brute Force FTP, Brute Force SSH, DoS, Heartbleed, Web Attack, Infiltration, Botnet, and DDoS, comprising over 2.8 million records in total. This dataset has become one of the most widely used benchmarks in the field of intrusion detection.

4.1.2. CIC-IoT2023 dataset

The CIC-IoT2023 benchmark dataset, released in 2023 by the Canadian Institute for Cybersecurity (CIC), is specifically designed to capture the inherent security challenges in modern Internet of Things (IoT) networks. By collecting attack traffic in a large-scale, realistic network environment, this dataset provides a standardized evaluation benchmark for research on detecting complex network threats.

The dataset is constructed based on an experimental topology involving 105 heterogeneous IoT devices, accurately simulating the dynamic interactions and security vulnerabilities in real IoT environments. All

Table 1

Description of the CIC-IoT-2023 dataset.

Attack category	Attack name	Total
DDoS	ACK Fragmentation	33,984,560
	UDP Flood	
	SlowLoris ICMP	
	Flood RSTFIN	
	Flood PSHACK	
	Flood HTTP	
	UDP Fragmentation	
	ICMP Fragmentation	
	TCP Flood	
	SYN Flood	
DoS	Synonymous IP Flood	8,090,738
	TCP Flood	
	HTTP Flood	
	SYN Flood	
	UDP Flood	
Mirai	GREIP Flood	2,634,124
	Greeth Flood	
Recon	UDPPPlain	354,565
	Ping Sweep	
	OS Scan	
	Vulnerability Scan	
	Port Scan	
Spoofing	Host Discovery	486,504
	Arp Spoofing	
Web-Based	DNS Spoofing	24,829
	Sql Injection	
	Command Injection	
	Backdoor Malware	
	Uploading Attack XSS	
Brute Force	Browser Hijacking	13,064
	Dictionary Brute Force	

attack samples in the CIC-IoT2023 dataset are generated by actual malicious IoT devices. By simulating adversarial interactions between attackers and target devices, the dataset presents highly realistic threat scenarios.

CIC-IoT2023 encompasses 33 fine-grained attack vectors categorized into seven main types: Distributed Denial of Service (DDoS), Denial of Service (DoS), Reconnaissance, Web-based Attacks, Brute-force, Spoofing, and Mirai Botnet attacks. The design of these scenarios ensures that the generated malicious traffic exhibits authentic characteristics in terms of protocol behavior, temporal features, and payload patterns. The dataset contains a total of 46,686,579 network events across 47 attributes, with benign traffic comprising only 2.35 % (1,098,195 records) of the entire dataset. A detailed breakdown of attack categories, sample counts, and percentage distributions is provided in Table 1.

4.2. Data preprocessing

The data preprocessing phase involves two major tasks to ensure the quality and balance of the dataset used for analysis. The first task is based on the attacker identification information provided in the CIC-IoT2023 dataset. We apply a precise filtering strategy based on attacker MAC addresses, adopting a bidirectional matching scheme: only traffic samples where the source or destination MAC address match entries in a known attacker address list are retained as valid attack samples. To avoid label contamination, any traffic associated with attacker addresses is strictly excluded from being classified as benign.

Through this process, we obtained a total of 45,897,804 valid records. To enable supervised learning, we assigned a label of 0 to benign samples (1,306,679 records) and 1 to malicious attack samples (35,607,434 records). To facilitate more fine-grained classification, we further divided malicious traffic into seven major categories, resulting in a total of eight labels including the benign class.

Table 2
Data distribution after data preprocessing.

Type	Flows after filter	Training data	Test data
Benign	1,306,976	1,045,581	261,395
DDoS	33,137,785	26,510,228	6,627,557
DoS	7,889,155	6,311,324	1,577,831
Mirai	2,568,491	2,054,793	513,698
Recon	931,805	745,444	186,361
Spoofing	55,807	44,646	11,161
Web-Based	5449	20,000	1976
Brute Force	2336	20,000	891

To improve the performance of deep learning models, we performed normalization of numerical features using the StandardScaler technique. The transformation follows Eq. (14):

$$\mathbf{x}' = \frac{\mathbf{x} - \mu}{\sigma} \quad (14)$$

where μ denotes the mean and σ denotes the standard deviation of each feature. This normalization ensures that all features follow a standard normal distribution with a mean of 0 and a standard deviation of 1, eliminating differences in scale and magnitude between features.

To address the class imbalance inherent in the CIC-IoT2023 dataset, we applied a combination of undersampling and oversampling techniques to create a balanced dataset. We split the dataset into training/validation and testing subsets, assigning 80 % of the data to training and validation. In the training set, we undersampled the majority classes and oversampled the minority classes to ensure each class is represented proportionally. The remaining 20 % is reserved as the test set. The distribution of filtered, training, and testing samples is shown in Table 2. This splitting strategy aims to enhance the generalization capability of the model.

An important aspect of our preprocessing procedure is ensuring balance across the subcategories of attacks. Since each main attack type in CIC-IoT2023 contains multiple subtypes, we ensure that these subtypes are proportionally represented in the training set. This not only preserves the diversity of the original dataset but also ensures that each attack variant is adequately represented, resulting in a more uniform data distribution and a representative sample space for training robust models.

4.3. Evaluation metrics

We employ four core metrics to evaluate the model's ability to distinguish between correct and incorrect predictions: (1) True Positive (TP), indicating the number of correctly classified benign samples; (2) False Positive (FP), denoting the number of attack samples incorrectly predicted as benign; (3) True Negative (TN), referring to the number of correctly classified attack samples; and (4) False Negative (FN), representing the number of benign samples incorrectly predicted as attacks. Based on these, we derive four evaluation metrics: Accuracy, Precision, Recall, and F1-score. Accuracy measures the proportion of correctly classified samples.

Precision measures the accuracy of correctly predicted benign samples, while recall assesses the model's ability to identify all benign samples. The F1-score, which is the harmonic mean of precision and recall, serves as a comprehensive metric for evaluating the performance of classification models. The formulas for these metrics are defined as follows.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (15)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (16)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (17)$$

Table 3
The settings for the hyperparameters in our experiment.

Hyperparameter	Value
Optimizer	Adam
Loss function	NLL
Batch size	64
Learning rate	$[10^{-2}, 10^{-5}]$
Epoch size	100
Hidden size	64
Attn size	32
Pooling method	Mean

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (18)$$

4.4. Experimental settings

The experiments were conducted using two NVIDIA RTX 3090 GPUs, each with 24GB of memory, in a Python 3.9 environment. The implementation is based on PyTorch 2.0.1 with CUDA 11.8, utilizing both PyTorch-Geometric [9] and DGL [30]. Model training employs the Adam optimizer, and the learning rate is dynamically adjusted via the ReduceLROnPlateau scheduler, with an initial range from 10^{-2} to 10^{-5} . A decay is triggered when the validation metric improves by less than 0.01 for five consecutive epochs. For a fair comparison, all baseline models are trained using the same hyperparameter configurations, as detailed in Table 3. This setup ensures computational efficiency while providing a stable training environment for all models.

4.5. Baseline methods

To validate the effectiveness of the proposed XMF-GNN model, we selected two single-modal models and three multi-modal models that combine traffic and packet-level information in some form for comparison. Additionally, we conducted experiments using several widely adopted machine learning models, including Random Forest (RF), Logistic Regression (LR), and K-Nearest Neighbors (KNN). We conducted a more comprehensive analysis of the experimental results on two datasets to comprehensively evaluate the effectiveness of our model. The following models are all advanced and representative baseline methods.

- These methods typically focus on a single aspect of network traffic, such as temporal or statistical characteristics, and employ deep learning or traditional machine learning algorithms for modeling. For instance, CNN-LSTM [26] uses a convolutional neural network (CNN) to extract spatial features from network traffic and a long short-term memory (LSTM) network to capture temporal dependencies, thereby modeling the spatiotemporal dynamics of traffic. The IARF model [22], based on an improved adaptive random forest, is specifically designed for detecting encrypted malicious traffic.
- The second category integrates flow-level features with packet payload characteristics to construct multimodal detection approaches. The GRL method [23] achieves bimodal traffic classification by fusing embedded representations of packet payloads with flow-level features. MultiModal NIDS [17] employs dual random forest models to process flow and payload information in parallel, enhancing detection accuracy. XG-NID [8] introduces a multimodal convolutional structure based on graph attention networks (GAT) to model the complex interactions between flow and payload from a graph-based perspective.

4.6. Experimental results and analysis

4.6.1. Overall performance analysis of XMF-GNN

To verify whether the proposed XMF-GNN model can achieve better malicious traffic detection capabilities, we compared it with five

baseline models and three widely used machine learning models. This section analyzes the results from two dimensions: multi-class anomaly detection and binary anomaly detection. Based on the experimental data, our model achieved the best metrics in both dimensions across two different datasets. These results provide empirical evidence for cybersecurity researchers selecting malicious traffic detection methods and also highlight the adaptive differences of various algorithm architectures across different detection scenarios.

In multi-classification experiments, the model must accurately identify the specific attack type to which network traffic belongs, which imposes higher representation and discrimination requirements compared to binary classification tasks. The experimental results are shown in Table 4. Overall, models with stronger multi-modal and graph structure modeling capabilities demonstrate a more significant advantage in multi-classification tasks. Traditional machine learning methods such as RF, LR, and KNN perform relatively poorly in multi-classification scenarios, particularly on the CICIDS-2023 dataset, where KNN's F1-score drops to 0.5191, highlighting its limitations in handling diverse attack behaviors. Meanwhile, most models perform better on the updated CICIDS-2023 dataset, indicating an improvement in dataset quality.

In contrast, deep learning models proposed in recent years, such as CNN-LSTM [26], GRL [23], and XG-NID [8], have demonstrated strong generalization and semantic modeling capabilities. Among these, XG-NID [7] achieved an F1-score of 0.9678 on CIC-IDS2017 and 0.9631 on CICIDS-2023, demonstrating robust overall performance. However, the proposed XMF-GNN model achieved the best or second-best performance in two multi-classification experiments: an F1-score of 0.9774 on CIC-IDS2017 and an even higher 0.9852 on CICIDS-2023, outperforming all comparison models.

The binary classification experiment simplifies the detection target into a binary classification of “normal traffic” and “attack traffic,” thereby avoiding the strong reliance of multi-classification methods on detailed attack labels. This approach demonstrates significant generalization advantages for unknown attack subtypes. The experimental results are shown in Table 5. The results indicate that, under both datasets, most models achieve higher accuracy in the binary classification task compared to multi-classification results. Our model still

demonstrates superior performance in the binary classification task when compared to these state-of-the-art methods. Overall, all models perform better in binary classification tasks than in multi-classification tasks. This is because binary classification tasks only require determining whether traffic is attack traffic, which is relatively simple; in contrast, multi-classification tasks require identifying specific attack types, placing higher demands on the model's representational capabilities. The single-modal models IARF and CNN-LSTM achieved accuracy rates of 0.9896 and 0.9989, respectively, but their accuracy decreased in multi-classification tasks, with IARF achieving an accuracy rate of 0.9537. This is because single-modal models often fail to capture the composite features of network traffic across temporal, structural, and semantic dimensions, leading to a lack of discriminative capability when faced with increasingly complex attack types.

Fig. 4 illustrates the model's training process. The model's accuracy on the training set rapidly increases to nearly 100 %, demonstrating its strong data fitting capability. As the number of iterations increases, the test accuracy remains slightly below the training accuracy, which may be due to the model learning excessive noise from some samples in the training set, thereby limiting its generalization ability. The sharp decline in the loss curve further confirms the model's rapid learning and convergence characteristics. However, when the loss value flattens out near 0.05, it indicates that the model's performance has already reached its peak before 50 epochs. Due to the implementation of a dynamic learning rate adjustment strategy, the model's learning rate has been continuously decreasing, which also indicates that our model has been converging.

Overall, the experimental results confirm that: (1) task complexity significantly affects detection performance, with binary classification tasks generally outperforming multi-classification tasks; (2) deep learning methods (especially multimodal and graph neural networks) outperform machine learning models in both types of tasks; (3) the XMF-GNN model we propose, based on heterogeneous graph and dynamic modality attention fusion mechanisms, significantly enhances the model's discriminative capability in multi-attack type recognition, demonstrating excellent test performance and reasonable generalization capabilities. It achieved the best performance in both datasets.

Table 4
Comparison of nine malicious traffic detection methods in multi-class scenarios. (Bold: best.)

Model	CIC-IDS2017				CIC-IDS2023			
	Accuracy	Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score
RF	0.7231	0.6237	0.8398	0.8130	0.8296	0.6437	0.6196	0.7437
LR	0.6281	0.6223	0.6389	0.6394	0.6589	0.6693	0.6641	0.6541
KNN	0.7542	0.7639	0.7747	0.7735	0.5442	0.5231	0.5902	0.5191
IARF	0.9336	0.9501	0.9661	0.9772	0.9537	0.9541	0.9682	0.9479
CNN-LSTM	0.9677	0.9600	0.9528	0.9413	0.9693	0.9689	0.9642	0.9611
MultiModal NIDS	0.8231	0.8342	0.8842	0.8295	0.8631	0.8638	0.8536	0.8532
GRL	0.9035	0.9325	0.9147	0.9235	0.9435	0.9562	0.9638	0.9665
XG-NID	0.9596	0.9574	0.9763	0.9678	0.9578	0.9665	0.9646	0.9631
XMF-GNN	0.9743	0.9736	0.9722	0.9774	0.9843	0.9863	0.9842	0.9852

Table 5
Comparison of nine malicious traffic detection methods in binary classification. (Bold: best.)

Model	CIC-IDS2017				CIC-IDS2023			
	Accuracy	Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score
RF	0.8231	0.8237	0.8598	0.8270	0.8696	0.8137	0.7996	0.8437
LR	0.8381	0.8223	0.8389	0.8394	0.8589	0.8693	0.8641	0.8641
KNN	0.8044	0.8031	0.7647	0.6435	0.8142	0.7831	0.7702	0.6991
IARF	0.9696	0.9637	0.9696	0.9631	0.9896	0.9836	0.9792	0.9828
CNN-LSTM	0.9723	0.9723	0.9692	0.9753	0.9989	0.9987	0.9693	0.9693
MultiModal NIDS	0.9402	0.9439	0.9442	0.9431	0.9842	0.9647	0.9645	0.9628
GRL	0.9472	0.9495	0.9562	0.9572	0.9972	0.9872	0.9835	0.9840
XG-NID	0.9855	0.9878	0.9895	0.9858	0.9965	0.9925	0.9929	0.9927
XMF-GNN	0.9872	0.9846	0.9883	0.9878	0.9983	0.9979	0.9980	0.9979

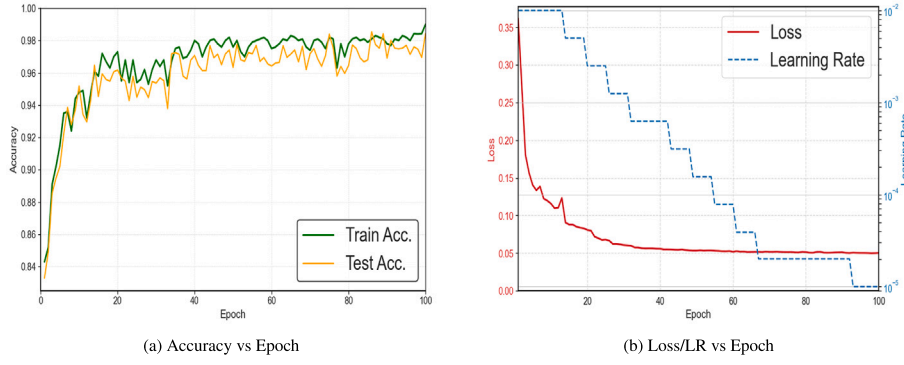


Fig. 4. Accuracy/Loss/LR vs Epoch.

4.6.2. Model performance comparison in multi-class malicious traffic detection

In multi-class classification experiments, to better focus on the performance of large categories, we only compared six deep learning models. We used weighted recall and weighted F1 scores to evaluate model performance. The experimental results are shown in Table 6. The experimental results on the CIC-IDS2023 dataset indicate that there are significant differences in detection performance among different models across eight traffic categories. Specifically, our model achieved optimal or near-optimal Recall and F1-score in many categories, particularly maintaining extremely high performance in challenging categories such as Benign, Mirai, Dos, DDoS, and BruteForce. This indicates that the attention mechanism we introduced effectively addresses the model's ability to perceive key features, avoiding the dilution of information from small-category samples, thereby enhancing the identification effectiveness of small-category attacks.

By comparison, although the XG-NID model achieves an F1-score of 1.000 in several categories, its recall is slightly lower in some cases. This may be due to its fusion strategy's heavy reliance on the dominant modality, causing the model to favor frequent classes under imbalanced data, thereby reducing its accuracy on rare classes. The GRL model performs well on WebBased and BruteForce categories but shows notable shortcomings on Mirai and DDoS attacks. This suggests that although GRL incorporates graph-based representation learning to enhance modality expression, it struggles with highly similar or high-dimensional heterogeneous data, where feature redundancy may undermine final classification results. While MultiModal NIDS is also a multimodal model, its performance lags significantly behind XMF-GNN and XG-NID, with consistently lower recall and F1-scores across most categories. This may be attributed to the lack of an effective coordination mechanism between modalities.

Among traditional single-modality models, CNN-LSTM and IARF show strong detection capability for large-scale attack types such as Mirai, DDoS, and Recon. However, their performance declines for small-sample or ambiguous attacks like WebBased and BruteForce. This

discrepancy reflects their strong modeling of temporal features but limited generalization ability.

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Confusion matrix. Fig. 5 shows the confusion matrix for six network intrusion detection models. The horizontal axis (Predicted label) represents the attack type label predicted by the model, i.e., the category into which the model classifies the input sample. The vertical axis (True label) represents the true attack type label of the sample, i.e., the actual attack category to which the sample belongs. The confusion matrix is an important tool for evaluating the performance of multi-classification models, as it visually displays the correspondence between the model's predicted results and the true labels in matrix form. To evaluate the multi-classification performance of the proposed XMF-GNN model, we conducted extensive experiments on the CIC-IDS2023 dataset and compared it with five baselines.

Table 6
Multi-class classification results for attack types. (Bold: best.)

Model	IARF		CNN-LSTM		MultiModal NIDS		GRL		XG-NID		XMF-GNN	
	Recall	F1-score	Recall	F1-score	Recall	F1-score	Recall	F1-score	Recall	F1-score	Recall	F1-score
Benign	0.9846	0.9026	0.9846	0.9726	0.8854	0.8509	0.8947	0.9354	0.9542	0.9813	0.9984	0.9876
WebBased	0.8641	0.8393	0.9365	0.9294	0.9351	0.8979	0.9381	0.9683	0.9913	0.8316	0.9973	0.9634
Spoofing	0.9635	0.9731	0.9742	0.9849	0.8741	0.8921	0.9368	0.9162	0.9264	0.9334	0.9876	0.9846
Recon	0.9542	0.9335	0.8421	0.9331	0.7423	0.7892	0.8402	0.9428	0.8317	0.9849	0.9341	0.9683
Mirai	0.8742	0.8879	1.0000	1.0000	0.8965	0.8717	0.8615	0.8491	1.0000	1.0000	1.0000	1.0000
Dos	1.0000	1.0000	0.9875	0.9906	0.9521	0.9362	0.9517	0.9643	0.9976	1.0000	1.0000	1.0000
DDoS	0.9979	0.9981	0.9624	0.9458	0.9263	0.8858	0.8962	0.8501	1.0000	1.0000	1.0000	1.0000
BruteForce	0.8422	0.8552	0.9687	0.9483	0.9567	0.9724	0.9841	0.9762	0.9771	0.9951	1.0000	1.0000

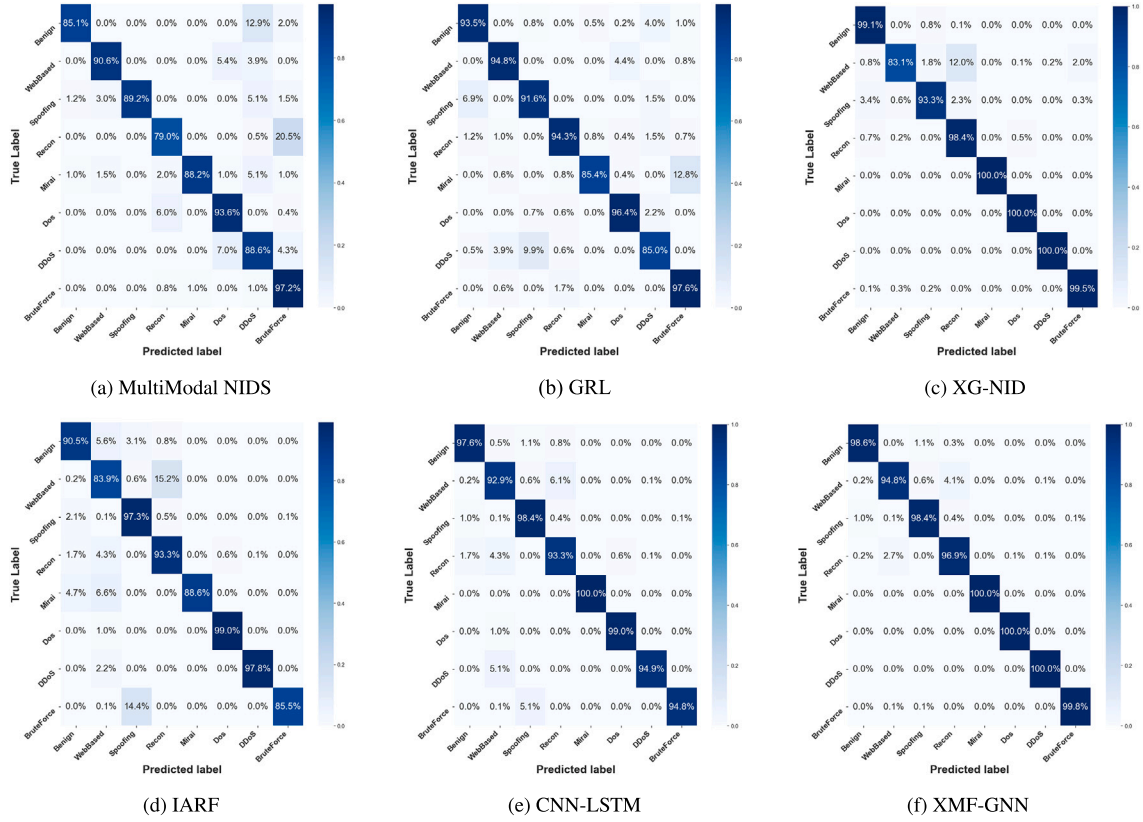


Fig. 5. Confusion matrices for six network intrusion detection models.

Fig. 5(a)–(c) and (f) show the MultiModal NIDS, GRL, XG-NID, and XMF-GNN methods, all of which are models that fuse multi-modal features. These models exhibit slightly better overall performance than single-modal models, but they still have many issues. MultiModal NIDS exhibits relatively low accuracy in identifying most attack types, which may be attributed to insufficiently refined fusion strategies, leading to inadequate learning in certain feature dimensions. GRL performs well on most attack types but has slightly higher misclassification rates for attacks like Mirai and DDoS, which may be due to insufficient cross-modal fusion or scarce training data. Although XG-NID achieves high accuracy on major attack types, its performance fluctuates significantly for anomaly-based attacks like Web-based and Spoofing, indicating that its generalization capabilities still have room for improvement.

In contrast, the traditional single-modal detection methods shown in Fig. 5(d) and (e) can effectively learn the main features of mainstream attacks and normal traffic, but their performance in handling small samples and detecting similar attack feature categories is significantly inferior to that of the fusion-based multi-modal detection model. IARF exhibits significant misclassification when detecting Web-based, Mirai, and BruteForce attacks, indicating its limitations in handling class imbalance and subtle attack patterns. The CNN-LSTM method performs excellently in identifying Benign, DoS, and Mirai attacks, but still exhibits detection biases in BruteForce and Recon types. This reflects its lack of cross-feature space expressive capability in temporal modeling, making it difficult to effectively distinguish attack samples with structurally similar but semantically distinct characteristics.

As shown in Fig. 5(f), the proposed XMF-GNN method demonstrates superior overall performance in multi-category network traffic detection, achieving an overall classification accuracy rate of nearly 99 %. The model achieves nearly perfect detection for most types of malicious traffic, demonstrating its strong ability to learn complex attack behavior features. This is attributed to its mechanism of integrating multi-modal

features and graph structure modeling, enabling the model to capture deep semantic connections across modalities.

Experimental results indicate that models incorporating multimodal information demonstrate improved performance compared to single-modal models in most classification tasks. However, due to the fixed fusion weighting scheme, these models cannot adaptively balance modalities based on the attack characteristics of the samples. In contrast, the proposed XMF-GNN method enhances the model's ability to model cross-modal semantic relationships by introducing graph structure modeling and multi-modal fusion mechanisms, thereby achieving higher detection accuracy and generalization performance in multi-classification tasks.

Case discussion on sample predictions. To further analyze the performance of the XMF-GNN model in multi-classification tasks, we selected some typical correctly predicted and incorrectly predicted samples based on the confusion matrix shown in Fig. 4(f) for case analysis, in order to gain a deeper understanding of the model's decision-making basis and potential biases in real-world scenarios. Each sample is derived from the test set corresponding to the model's prediction results, with the sample ID indicating its index position within the graph neural network's input batch, ensuring traceability. For example, sample ID “#001” represents the first graph structure sample used for testing in the dataset, including its corresponding flow nodes, packet nodes, and their features. By analyzing the key features of these samples, we can gain a deeper understanding of the model's decision-making criteria and identify input patterns or feature combinations that may lead to misclassifications, thereby providing targeted directions for model optimization.

Although the overall classification accuracy of the XMF-GNN method we propose is close to 99 % in multi-category network traffic detection, there are still a small number of misclassified samples, as shown in Table 7. From the correctly predicted samples (#001, #017, #035),

Table 7
Correct/incorrect prediction samples.

Sample ID	Actual	Predicted	Key features
#001	BruteForce	BruteForce	pkt_count_60s byte_entropy (flow)
#017	Mirai	Mirai	payload_length byte_std (packet)
#035	Dos	Dos	std_iat pkt_rate_10s (flow)
#062	WebBased	Recon	conn_rate ttl_var payload_len (flow)
#088	Benign	Spoofing	ttl src_port_var byte_entropy (flow)

it can be seen that XMF-GNN can accurately capture key indicators with attack characteristics, such as `pkt_count_60s`, `payload_length`, and `std_iat`, and through the joint modeling of Flow and Packet features, achieve effective discrimination of attack types. These features typically exhibit significant deviations at the statistical level (e.g., transmission frequency, entropy, and interval), reflecting the model's sensitivity and judgment of abnormal behavior. For example, BruteForce is characterized by extremely high packet counts and high-entropy byte distributions, indicating strong brute-force attack features, enabling the model to accurately identify such attacks. DoS attacks are characterized by extremely small interval standard deviations and high-frequency packet transmission within a short timeframe, with feature combinations clearly pointing to DoS attacks.

Additionally, in the misclassified samples (#062, #088), some Benign or Web-based traffic exhibits statistical behaviors similar to Recon or Spoofing (such as TTL fluctuations or port changes) under specific conditions, thereby misleading the model. This is due to high connection rates and TTL fluctuations in Web-based features causing misclassification, which resemble Recon detection features, lacking deeper semantic recognition. Frequent port switching and fixed low TTL values cause the model to misclassify a small portion of benign samples as spoofing. This indicates that the current model still has ambiguous discrimination for attack boundaries with weak semantics but statistically similar characteristics. In the future, it may be considered to introduce deep protocol behavior features, content structure modeling, or contextual semantic information to further enhance the model's robustness and discrimination accuracy in complex attack scenarios.

4.6.3. Ablation study

To validate the impact of the proposed dynamically attention-driven modality fusion module on the XMF-GNN model, we conducted a series of ablation experiments. Each experiment was repeated multiple times to minimize errors, and the average results and time efficiency are presented in Table 8. The baseline model, B-GNN, constructs a graph solely based on flow features and performs intrusion detection without introducing any modality fusion mechanism. It serves as a benchmark to assess the performance improvements brought by the subsequent fusion modules.

On top of B-GNN, FM-GNN incorporates a simple modality fusion module that uses an MLP to perform weighted fusion of flow and packet features into a unified representation for graph modeling and classification. GA-GNN introduces a gating mechanism to enhance controllability over attention distribution. SA-GNN applies a structurally simpler attention mechanism, where modality fusion weights are computed through a learnable linear layer. MA-GNN extends FM-GNN by integrating a multi-head attention mechanism, which enables parallel learning of diverse fusion patterns via multiple attention subspaces.

As shown in Table 8, removing any module results in a decrease in malicious traffic detection performance, with the dynamic attention mechanism playing a pivotal role among all modules. After introducing the dynamic attention mechanism, detection accuracy improves significantly, which can be attributed to its ability to adaptively assign weights to flow and packet features based on their modal embeddings. This allows the model to more accurately capture the semantic preferences of different attack types.

Table 8
Impact of different attention mechanisms on model performance.

Model	Accuracy	Time cost (s)		
		Training process	Detecting process	Overall
B-GNN	0.9578	135.64	4.32	139.96
FM-GNN	0.9759	181.87	5.21	187.08
SA-GNN	0.9792	231.45	7.41	238.86
GA-GNN	0.9810	354.77	10.62	365.39
MA-GNN	0.9840	1309.33	160.54	1469.87
XMF-GNN	0.9843	259.45	11.45	270.90

We also observe that adding multi-head self-attention significantly improves accuracy over the baseline model. This is because multi-head attention can flexibly learn the associations and importance among features, allowing the model to assign appropriate weights according to task requirements. However, incorporating multi-head attention also leads to a substantial decrease in time efficiency, and the accuracy gain is relatively marginal. This suggests that although this mechanism enhances expressive power in feature fusion, the attention heads tend to stabilize at fixed weight distributions over time, indicating limited dynamic adaptability. In contrast, the dynamic attention mechanism shows better generalization performance.

4.6.4. Node feature analysis

To gain a deeper understanding of how the XMF-GNN model utilizes the features of different nodes to make predictive decisions in classification tasks, we employed the integrated gradient (IG) method to conduct an importance analysis of the features of Flow nodes and Packet nodes. The `torch_geometric.explain` module provided by PyTorch-Geometric offers a specific implementation method. The specific results are shown in Figs. 6 and 7. This method effectively measures the contribution of each input feature to the model's final output, making it one of the commonly used attribution techniques in current neural network interpretability research.

IG is a gradient explainability method based on path integrals. Unlike traditional gradient methods, IG obtains more stable and comprehensive feature importance scores by smoothly transitioning input features from a "baseline value" to actual input values and calculating the integral of the gradient along this path. This method avoids gradient saturation issues and has strong theoretical explainability.

For an input feature vector $\mathbf{x} \in \mathbb{R}^d$, let the model be $F(\mathbf{x})$, and the baseline input be $\mathbf{x}'$. The integral gradient calculation formula for the i -th feature is as follows:

$$IG_i(\mathbf{x}) = (x_i - x'_i) \cdot \int_{\alpha=0}^1 \frac{\partial F(\mathbf{x}' + \alpha(\mathbf{x} - \mathbf{x}'))}{\partial x_i} d\alpha \quad (19)$$

Here, $\mathbf{x}$ is the target input, $\mathbf{x}'$ is the baseline input, F is the model's prediction function, $\frac{\partial F}{\partial x_i}$ denotes the gradient with respect to the i feature, and $IG_i(\mathbf{x})$ represents the integrated gradient contribution of feature i to the model's output.

In the attribution analysis of Flow node features (Fig. 6), the model clearly relies more on statistical and time-series features, with multiple features scoring above 0.75 in importance, indicating their central role in model decision-making. Among these, `flow.duration` (flow duration), `flow.packets` (total number of packets), and `flow.bytes` (total number of bytes) achieved attribution scores of 0.90, 0.88, and 0.85, respectively, reflecting sensitivity to overall traffic intensity. The statistical features extracted under the multi-scale sliding window and the global traffic statistical features together form the key basis for the model's predictions, validating the effectiveness of the multi-scale feature extraction module designed in Section 3.1. For example, `byte_sum_30s` scored 0.82 and `pkt_count_60s` scored 0.77, indicating that the model effectively leverages traffic distribution patterns within time windows to identify periodic or high-frequency attacks such as DDoS and BruteForce. Other

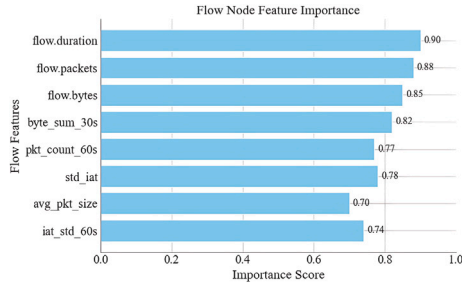


Fig. 6. Flow feature vector.

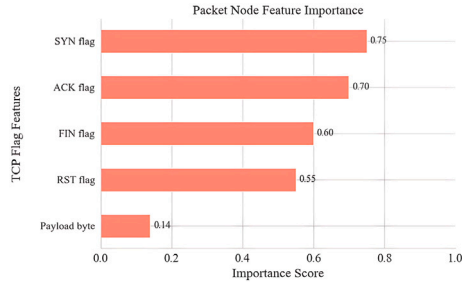


Fig. 7. Packet feature vector.

features such as `std_iat` and `iat_std_60s` (packet interval time and its standard deviation) also achieved high scores, indicating that the model has strong modeling capabilities for traffic time behavior, which helps distinguish normal communication from scanning or probing attacks.

In the Packet node features (Fig. 7), the model primarily relies on TCP flags to determine attack types. The scores for the syn and ack features

are 0.75 and 0.70, respectively, indicating that the frequency or pattern of SYN and ACK packets has high discriminative power in classification tasks, which is critical for identifying specific attack types such as SYN Flood and BruteForce. In contrast, the scores for the fin and rst features are slightly lower at 0.60 and 0.55, respectively, suggesting their limited discriminative power in specific attack scenarios. It is worth noting that the payload (payload byte sequence) feature score is only 0.14, indicating that the model relies weakly on the original payload content. This may be due to its high redundancy or significant noise, making it unable to consistently provide discriminative information. This also reflects that the model prefers to judge anomalies based on control flags and behavioral patterns rather than byte-level semantic content.

Fig. 8 illustrates the variation trends of flow modality attention weights during training across different models. As shown in Fig. 8(d), at the beginning of training, parameters are not yet converged and the value of each modality has not been effectively distinguished, resulting in exploratory behavior by the attention mechanism. In the middle phase of training, the weights fluctuate between 0.60 and 0.67 and show an upward trend overall, indicating that the model is gradually learning to “recognize” the more informative modality. In the late phase, the weights stabilize around 0.65 with minimal fluctuations, demonstrating that the fusion mechanism has converged. While flow features dominate the final classification, the presence of other attack types relying more on packet-level information prevents full suppression.

Fig. 8(a) shows that despite its structural complexity, the multi-head attention mechanism degenerates into a static weighting mechanism during training, failing to fully utilize the complementary capabilities of multiple heads. In contrast, Fig. 8(b) and (c) clearly demonstrate that the dynamic attention mechanism in our model exhibits faster convergence in capturing modality dependency differences among attack types and achieves better stability.

The experimental results show that the dynamic attention mechanism significantly enhances the model’s ability to detect flow-behavior-driven attacks (e.g., DDoS, Mirai), while maintaining high precision in

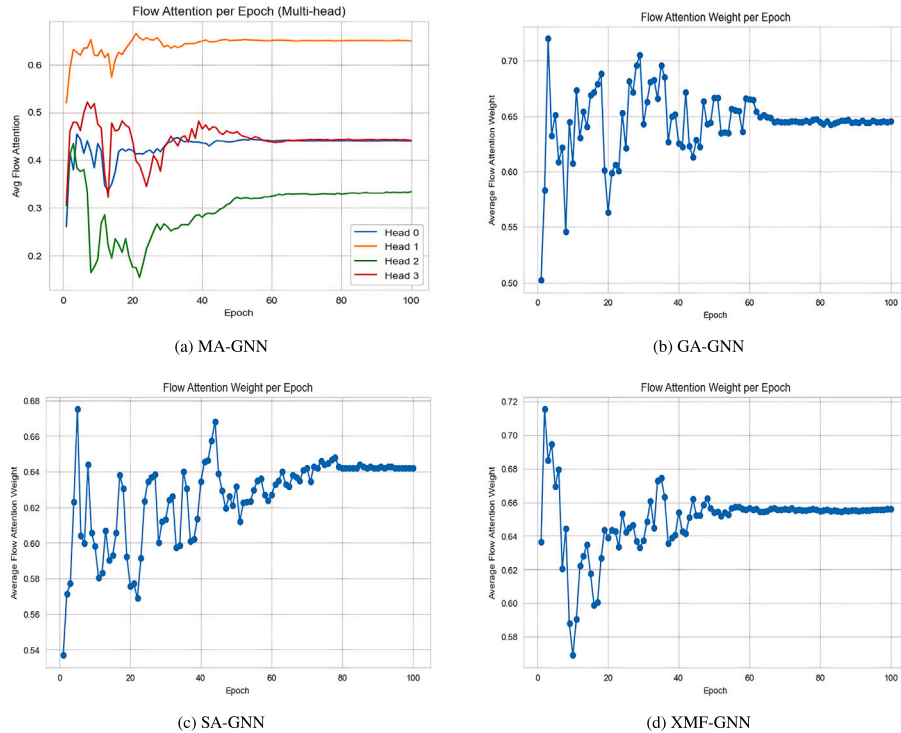


Fig. 8. Variation trends of flow-modality attention weights during four attention training processes.

detecting payload-driven attacks (e.g., BruteForce, WebBased). This improves the intrusion detection system's adaptability and generalization capability across diverse types of cyberattacks.

5. Conclusion

This paper proposes a cross-modal dynamic fusion-based heterogeneous graph neural network model (XMF-GNN) to address key challenges in current network intrusion detection, such as insufficient utilization of multi-modal information and poor adaptability of static fusion mechanisms. The model constructs heterogeneous graph structures at the flow level (Flow) and packet level (Packet), and introduces an attention-driven dynamic fusion mechanism to enable adaptive weight learning across modalities, thereby significantly enhancing the model's ability to distinguish between various attack patterns. Experimental results show that XMF-GNN performs excellently on two different datasets, achieving an F1-score of 98.4 % in the multi-class classification task on the CIC-IDS2023 dataset, fully validating the model's excellent generalization and superiority.

Despite the good results, this study still has certain limitations. First, although the dynamic fusion mechanism of XMF-GNN demonstrates good adaptability during training, it may still affect the model's convergence speed and the stability of the final discrimination boundary in scenarios where node feature distributions are highly sparse or subject to strong noise interference. Additionally, the model's long-term robustness and adaptability in the face of data distribution shifts or evolving real-world attack behaviors require further optimization.

Future research will focus on the following areas: first, expanding to more multimodal intrusion detection datasets to systematically evaluate its cross-scenario adaptability; second, combining graph-enhanced learning with adversarial training mechanisms to enhance the model's robustness under extreme conditions such as sparse graph structures, high abnormal noise, or missing nodes; third, further exploring the cross-domain transferability and generalization capabilities of XMF-GNN, such as applying it to fraud detection in e-commerce platforms or anomaly transaction identification in banking and financial networks. Since these domains exhibit heterogeneity in attack behaviors, network structures, and feature spaces compared to IoT, future research may leverage domain adaptation and knowledge distillation techniques to achieve more efficient cross-domain transfer.

Finally, the interpretability advantage of XMF-GNN also provides a solid foundation for its deployment in real-world systems. In the next phase, we plan to integrate this model into an automated security platform with online learning and response capabilities, enabling intelligent, efficient, and autonomous network intrusion response and decision support, and further advancing the practical application of multimodal graph intelligence in cybersecurity.

CRedit authorship contribution statement

Zhengxiang Ma: Writing – original draft, Funding acquisition, Formal analysis, Conceptualization. **Yan Liu:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Conceptualization. **Yu Chen:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Data curation. **Zhibin Liu:** Writing – original draft, Software, Methodology, Formal analysis. **Yinghao Li:** Writing – review & editing, Writing – original draft, Validation, Investigation, Conceptualization.

Declaration of competing interests

The authors declare the following financial interests/personal relationships which may be considered potential competing interests:

Zhengxiang Ma reports that financial support was provided by The Key Research and Development Program of Henan Province (No. 231111211500, No. 241111223000). If there are other authors, they declare that they have no known competing financial interests or personal

relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

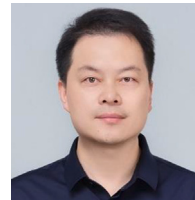
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Yan Liu obtained a bachelor's degree in network engineering from Xinyang Agricultural and Forestry University in China in 2022. He is currently pursuing a master's degree in artificial intelligence at Zhengzhou University of Aeronautics in China. He holds several national patents and competition awards related to cybersecurity. His research interests include network and information security, artificial intelligence, and deep learning.



Yu Chen received the M.S. degree in circuit and system from Northwestern Polytechnical University, Xi'an, China, in 2009, and the Ph. D. degree in electronic science and technology from Northwestern Polytechnical University, Xi'an, China, in 2017. His research interest include data acquisition and signal processing, network information security, navigation guidance and control, and network collaborative manufacturing.



Zhibin Liu received his bachelor's degree in 2022 and is currently studying in the Department of Electronic Information at Zhengzhou University of Aeronautics in China. His main research directions are natural language processing and deep learning. He is committed to researching information extraction technology.



Yinghao Li received the B.S. degree in computer science and technology from the University of Zhengzhou University, China, in 2010, and the Ph.D. degree in computer science and technology from the University of Chongqing University, China, in 2016. He is currently an associate professor at the Department of School of Cyber Science and Engineering, Zhengzhou University. His main research interests include machine learning, pattern recognition, and image processing.

Author biography



Zhengxiang Ma obtained a master's degree in industrial engineering from Tianjin University (Tianjin, China) in 2009 and and the Ph.D. degree in control theory and control engineering from Zhengzhou University (Zhengzhou, China) in 2016. He is a professor at Zhengzhou University of Aeronautics (Zhengzhou, China), a recipient of a special government allowance from the State Council, and a leading academic and technical expert in Henan Province. He has long been engaged in engineering technology and applied research in the fields of information systems, electronic intelligence, and network engineering.